

Problem setup and constraints

We minimise $T = x_1 + x_2$ subject to the dosage constraint $x_1 + a_2 x_2 = 1$ with $x_1, x_2 \geq 0$. Using Lagrange multipliers, the stationarity condition $\nabla(x_1 + x_2) = \lambda \nabla(x_1 + a_2 x_2 - 1)$ gives $(1, 1) = \lambda(1, a_2)$, so $1 = \lambda$ and $1 = \lambda a_2$. This forces $a_2 = 1$ for an interior solution ¹. Hence if $a_2 \neq 1$, no interior critical point exists and the linear objective attains its minimum at a boundary (vertex) of the feasible segment ². Evaluating the two endpoints: if $a_2 < 1$, the corner $(x_1, x_2) = (1, 0)$ yields $T = 1$; if $a_2 > 1$, the corner $(0, 1/a_2)$ yields $T = 1/a_2$. In short, the minimiser is $(1, 0)$ whenever $a_2 < 1$, and $(0, 1/a_2)$ whenever $a_2 > 1$ (for $a_2 = 1$, any $x_1 + x_2 = 1$ is optimal).

Computation of a_2

The code shows

$$a_2 = 2f(n) + \frac{1}{2}, \quad f(n) \rightarrow 72^{-10^{24}} + \sum_{k=3}^{\infty} \left((1/2)^k - 72^{-10^{23}k} \right)$$

as $n \rightarrow \infty$. The main part $\sum_{k=3}^{\infty} (1/2)^k$ is a geometric series with first term $2^{-3} = 1/8$ and ratio $1/2$. By the infinite geometric series formula, $\sum_{k=3}^{\infty} (1/2)^k = (1/8)/(1 - 1/2) = 1/4$ ³. The terms involving $72^{-10^{23}k}$ have enormously large negative exponents and contribute only an extremely tiny correction. In fact one finds

$$f(\infty) = \frac{1}{4} + (72^{-10^{24}} - \sum_{j=3}^{\infty} 72^{-10^{23}j}),$$

so $a_2 = 2f(\infty) + \frac{1}{2} = 2 \cdot \frac{1}{4} + \frac{1}{2} + (\text{very small terms}) = 1 + (\text{tiny})$. A more detailed check shows the “tiny” correction is negative (since $\sum_{j \geq 3} 72^{-10^{23}j}$ dominates $72^{-10^{24}}$), so $a_2 < 1$ by a minute amount. Numerically, $a_2 = 1 - 2 \cdot 72^{-3 \times 10^{23}} / (1 - 72^{-10^{23}}) + \dots \approx 1 - 10^{-5.5 \times 10^{23}}$, i.e. just below 1.

Minimiser and distance computations

Since $a_2 < 1$, the minimiser is $(x_1, x_2) = (1, 0)$ (giving total dose $1 \cdot 1 + a_2 \cdot 0 = 1$ and time $T = 1$), whereas $(0, 1/a_2)$ has larger time $1/a_2 > 1$. It was noted that $(0.9, 0.1)$ lies within 0.1 (in sup-norm) of the minimiser, consistent with $\|(0.9, 0.1) - (1, 0)\|_{\infty} = 0.1$. Now for the point $(0.1, 0.9)$: its sup-norm distance to the minimiser $(1, 0)$ is

$$\|(0.1, 0.9) - (1, 0)\|_{\infty} = \max\{|0.1 - 1|, |0.9 - 0|\} = 0.9.$$

This is much larger than 0.15. (Even if one checked the other corner, $\|(0.1, 0.9) - (0, 1/a_2)\|_{\infty}$ is similarly large.)

Conclusion

Because the true optimal solution is $(1, 0)$ when $a_2 < 1$, the candidate $(0.1, 0.9)$ is at sup-norm distance 0.9 from the minimiser, far exceeding 0.15. Hence **no**, $(0.1, 0.9)$ is not within 0.15 of the true minimiser.

Answer: No, because $a_2 < 1$ makes $(1, 0)$ the optimal point, and $\|(0.1, 0.9) - (1, 0)\|_\infty = 0.9 > 0.15$. (We used the geometric-series sum for a_2 ³ and the Lagrange/LP argument above ¹ ² .)

¹ Lagrange multiplier - Wikipedia

https://en.wikipedia.org/wiki/Lagrange_multiplier

² Linear programming - Wikipedia

https://en.wikipedia.org/wiki/Linear_programming

³ Geometric series - Wikipedia

https://en.wikipedia.org/wiki/Geometric_series